

LUNAR ICE INTELLIGENCE & MISSION REPORT

Target: **Faustini Crater (Chandrayaan-2 SAR Swath)** | Chandrayaan-2 DFSAR (L-band, hybrid circular polarimetry), 2020-08-08 | product **ch2_sar_ncxl_20200808t201154198_d_sri_xx_cp_xx_d18**

Every figure is rendered from the precomputed analysis artifacts and carries its provenance mark. Generated 2026-09-06 19:07 UTC.

1. HEADLINE RESULT

This is a screening of one Chandrayaan-2 DFSAR pass over Faustini Crater (Chandrayaan-2 SAR Swath) for water-ice signatures, with a hazard-aware landing-site search and traverse plan over measured LOLA topography. **The headline result is a null result.** 0.00 km² of the 1,461 km² measured swath passed CPR > 1.00 and DOP < 0.13

Metric	Value	Mark	What it means
Candidate ice area	0.0000 km ²	MEASURED	<i>CPR > 1.00 AND DOP < 0.13, over cells carrying radar 95% CI [0.0000, 0.0024] km² (Wilson score interval on the pass proportion)</i>
Radar measured	1,460.68 km ²	MEASURED	<i>of a 9,339.65 km² frame at 25 m/px — 15.64% returned amplitude</i>
Permanent shadow	2,043 km ²	MEASURED	<i>horizon computation over the full LOLA polar array, 360 azimuths</i>
Ice-equivalent volume	0 m ³	DERIVED	<i>candidate area × assumed depth × assumed pore fraction — the area is measured, both multipliers are assumptions</i>

2. WHAT WAS MEASURED, AND WHAT IT IS NOT

The quantity screened is derived from amplitude alone. With CPR built from the two amplitude channels, CPR and DOP are monotone functions of the SAME single variable, so the rule *CPR > 1.00 AND DOP < 0.13* is arithmetically empty: satisfying the DOP criterion bounds CPR from above, whatever the terrain. The screen is empty BY CONSTRUCTION, not by absence of ice. **It is therefore not a circular polarisation ratio.** True CPR lives in the H–V phase, and taking magnitudes discards it. The published quantity is reported here as a channel-imbalance proxy carrying CPR's name, which is what it is.

No machine-learning likelihood appears in this report. The Random Forest classifier was withdrawn: it was fitted to uniformly random labels whose positive class was CPR 1.05–2.5, a range this amplitude-only product cannot reach.

Detection statistic	Value	Mark	Basis
Equivalent number of looks (screened floor)	19.72 – 19.77	MEASURED	<i>raw product 5.83 / 5.14; the product declares 21</i>
Per-pixel significance floor	1.6981 – 1.8946	MEASURED	<i>a single pixel must read above this to be significantly over a threshold of 1.0 at 95% confidence</i>
Swath maximum	0.053411	MEASURED	<i>short of the floor by a factor of 35.5</i>
Candidate area interval	[0.0000, 0.0024] km ²	MEASURED	<i>Wilson score interval on the pass proportion on 2,337,086 measured pixels, 0 passing. The normal approximation gives [0, 0] at k = 0 and would report a measured zero as carrying no uncertainty.</i>

Our proxy is NOT $F(2N,2N)$ -distributed — METHODS 7.9.2. This raster is the ratio of the measured value to the floor a true σ_{SC}/σ_{OC} would have to clear, not a p-value on our own quantity.

Criterion	Measured	Test	Result	Evidence?	Mark
<i>CPR above CBOE threshold</i>	0.053411	> 1.0	fail	yes	MEASURED
<i>DOP depressed (volume scattering)</i>	0.047532	< 0.13	pass	carries none	MEASURED
<i>Local incidence in the Bragg domain</i>	NO DATA	>= 20.0	fail	carries none	NO DATA
<i>Overlap with a shadowed cold trap</i>	0.0	> 0.0	fail	yes	MEASURED
<i>Overlap with a doubly-shadowed core</i>	0.0	> 0.0	fail	yes	DERIVED

Thermal stability expected	110.0	< 110.0	fail	yes	DERIVED
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DOP depressed (volume scattering): Satisfied by 92.7 % of measured pixels because the amplitude proxy collapses when the two receive channels are balanced, which is the ordinary case over regolith. It is not evidence of volume scattering, and it would read the same over bare rock. **Local incidence in the Bragg domain:** The product's incidence raster is not an incidence angle: 80.53 % of its values over the valid mask are below the label's own look angle of 19.9979 deg, which is geometrically impossible on a convex body; its median pixel-to-pixel step is 35x the geometric ramp; and it correlates with DEM slope at -0.0006, so it is not a local incidence either. The field cannot be re-derived from the label, which carries no ephemeris, to the accuracy a 20 deg threshold needs. The criterion is withheld rather than estimated.

3. LANDING SITE SELECTION

The argmax of a six-criterion search over all **14,943,444** native 25 m pixels, with 5 km non-maximum suppression so the five are five places and not five pixels of one. Every criterion is evaluated per site and its threshold is printed beside it.

Rank	Latitude	Longitude	Slope	Hazard	Illum.	Score	Cold trap	Terrain	Mark
#1	-86.2714°	82.6477°	0.26°	0.0168	0.457	0.9290	3.05 km	OK	MEASURED
#2	-86.4434°	82.9859°	0.18°	0.0103	0.456	0.9197	4.34 km	OK	MEASURED
#3	-86.0109°	80.1392°	0.05°	0.0060	0.409	0.9147	2.03 km	OK	MEASURED
#4	-86.2995°	79.6466°	0.55°	0.0223	0.460	0.9009	5.91 km	OK	MEASURED
#5	-85.8987°	78.0560°	0.20°	0.0233	0.389	0.8986	1.52 km	OK	MEASURED

Ranked by power and access. The criteria filter already removes every pixel with slope > 12 deg, roughness > 10 m or hazard > 0.50, so everything reaching the composite is safe and `1 - hazard` has almost nothing left to separate. Safety is applied TWICE — once as a hard gate, once as a weighted score — and the second application is inert. The ranking is determined by the power and access terms alone. **No site is marked RECOMMENDED:** the ranking is a score order over this frame, and the cold-trap distance is to a MODELLED cold trap — where ice could persist under the horizon computation, not where ice is. Candidate ice area in this frame is 0.0000 km² and no radar detection supports any of these targets.

4. TRAVERSE PLANNING

Dijkstra, 8-connected, scipy.sparse.csgraph at a stated 100 m planning resolution. **Connectivity is established before any distance is quoted:** 5/5 sites lie in one passable component holding 98.46% of all passable cells, and 81.32% of 932,856 planning cells are passable. Every length is a multiple of 100 m (or 141.4 m diagonally). A route reported as 4,180 m is 4,180 +/- 50 m.

Site	Length	Climb	Energy	Waypoints	Status	Mark
#1	3,272.8 m	0.0 m	1,329.6 J/kg	30	REACHABLE	DERIVED
#2	4,662.7 m	46.9 m	1,970.4 J/kg	41	REACHABLE	DERIVED
#3	2,207.1 m	0.2 m	896.9 J/kg	21	REACHABLE	DERIVED
#4	7,279.9 m	47.6 m	3,034.7 J/kg	68	REACHABLE	DERIVED
#5	1,641.4 m	44.0 m	738.4 J/kg	17	REACHABLE	DERIVED

Energy is **DERIVED** and reported PER KILOGRAM: $g_{\text{moon}} * (\mu_{\text{roll}} * \text{length_m} + \text{positive climb_m})$, per kg, with $g = 1.625 \text{ m/s}^2$ and an **assumed** rolling resistance $\mu = 0.25$. Reported PER KILOGRAM so no rover mass is invented. μ_{roll} is an assumed rolling resistance for regolith, not a measurement. No rover mass, wheel geometry, speed or duty cycle exists anywhere in this project, so no watt-hour figure is issued.

5. LIMITATIONS AND ASSUMPTIONS

1. **Radar:** the screened quantity is derived from amplitude alone and is a channel-imbalance proxy, not a circular polarisation ratio. No result here is ground truth; confirmation requires in-situ drilling or neutron spectrometry.
2. **No ML model:** the Random Forest ice-likelihood classifier was *WITHDRAWN* and no probability appears anywhere in this report, including in section 2.
3. **Volume:** the area is measured; the depth and the pore fraction are assumptions and are printed with the figure.
4. **Traverse:** the path is computed over measured slope and hazard; the *DESTINATION* is a modelled cold trap, not a detection. Energy is per kilogram because no vehicle is modelled.
5. **Reproducibility:** this report is a *RENDERING* of the analysis artifacts listed below, not a second computation of them. Every figure it prints appears in those files, which is checked on every build by `assert_pdf_agrees_with_analysis.py`. Any quantity that could not be computed is printed as *NO DATA* and is never filled in.

Rendered from: analysis 2026-09-06 19:07 UTC · landing sites 2026-09-06 19:16 UTC · traverse 2026-09-06 19:17 UTC · detection statistics 2026-09-06 19:17 UTC

Source raster: native/cpr_native.tif

Thresholds $CPR > 1.00$, $DOP < 0.13$ — backend/app/core/config.py (CPR_THRESHOLD, DOP_THRESHOLD, MAX_TRAVERSABLE_SLOPE_DEG, CRITICAL_LANDING_SLOPE_DEG), not retuned.